REAL ANALYSIS



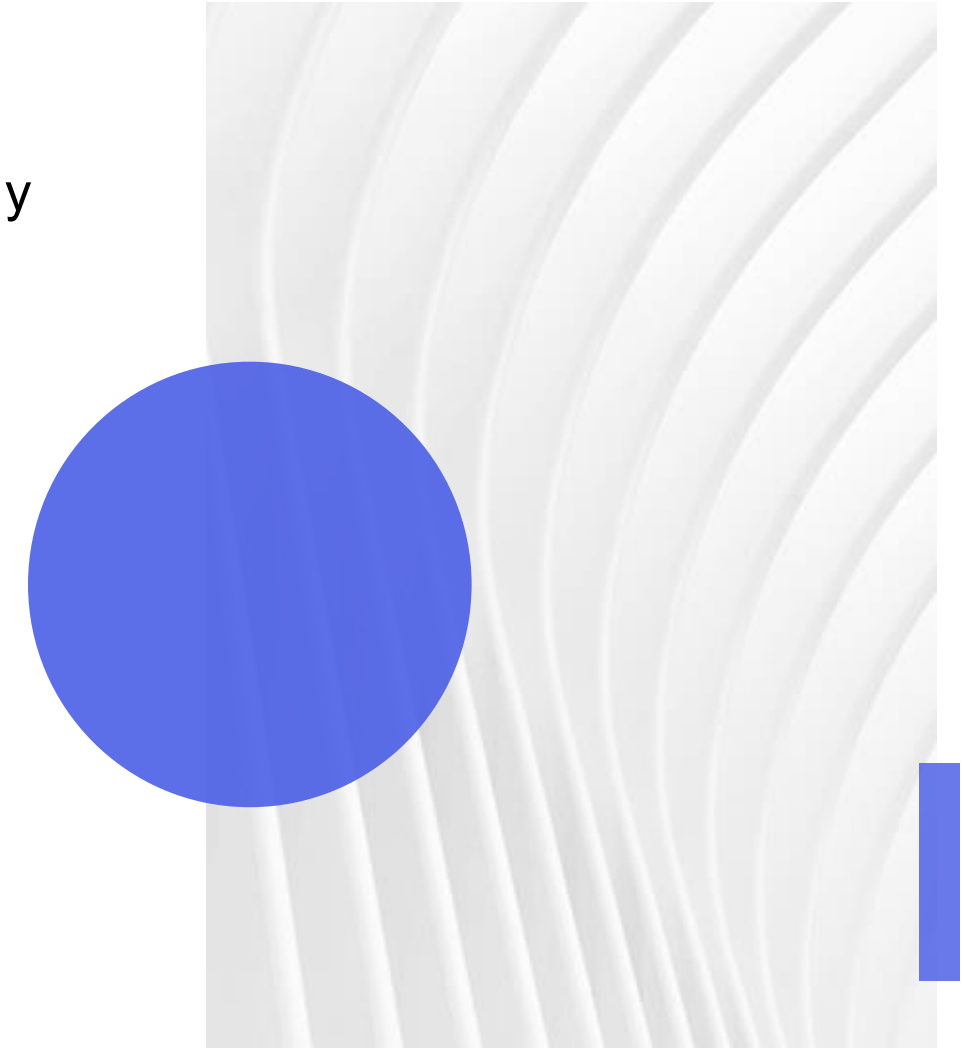
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Real Analysis Club

https://www.youtube.com/channel/UCdXWeu5wFkjCJejVYU2MzDQ

Fields

A field is a set $\mathbb{F}$ on which are defined operations of addition and multiplication, associating elements x + y and xy of $\mathbb{F}$ to each pair x, y of elements of F, for which the following axioms are satisfied:

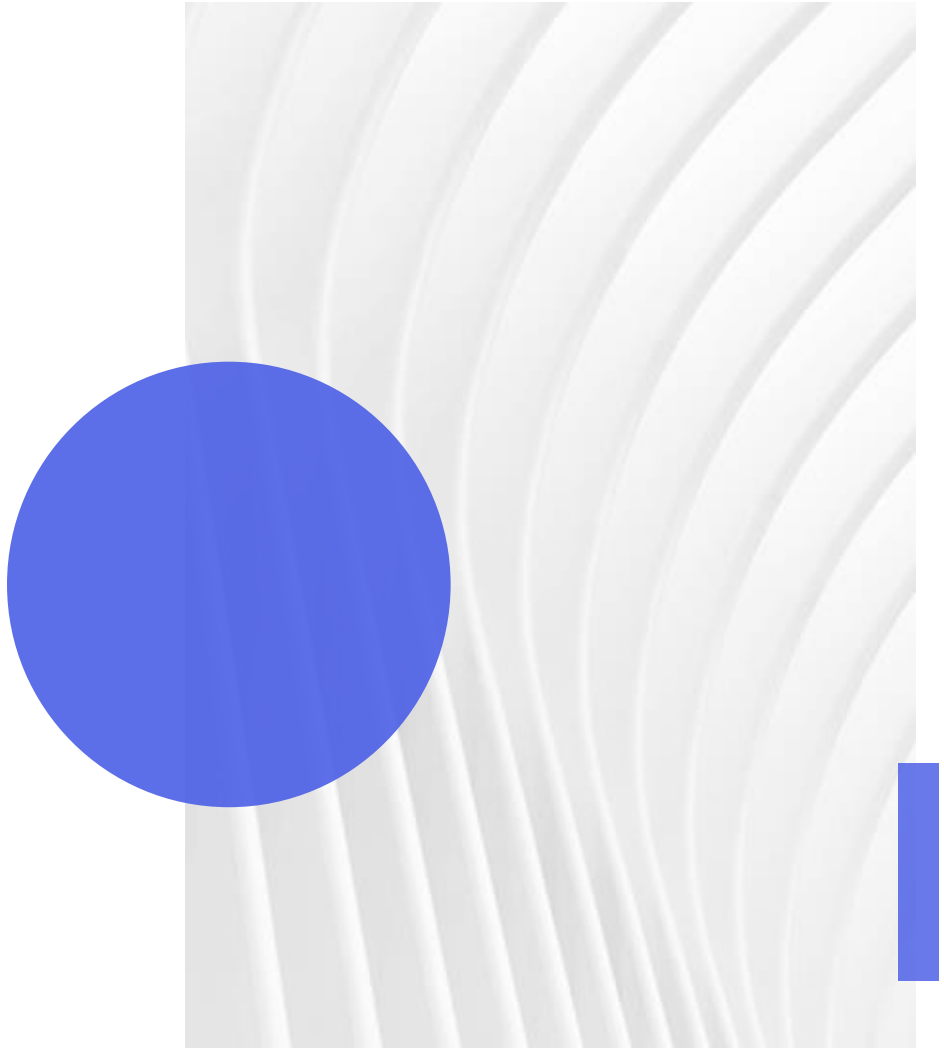
1. the operation of addition on $\mathbb{F}$ is commutative.
2. the operation of addition on $\mathbb{F}$ is associative.
3. there exists a zero element for the operation of addition on $\mathbb{F}$.
4. negatives of elements of $\mathbb{F}$ always exist.
5. the operation of multiplication on $\mathbb{F}$ is commutative.
6. the operation of multiplication on $\mathbb{F}$ is associative.
7. there exists an identity element for the operation of multiplication on $\mathbb{F}$.
8. given any x $\in$ $\mathbb{F}$ satisfying x $\neq$ 0, there exists an element x^{-1}. of $\mathbb{F}$ satisfying x $x^{-1} = 1$.
9. multiplication is distributive over addition.



Fields

What about subtraction and division

The operations of subtraction and division are defined on a field $\mathbb{F}$ in terms of the operations of addition and multiplication on that field in the obvious fashion: $x-y = x + (-y)$ for all elements x and y of F, and moreover $x/y = xy^{-1}$ provided that $y \neq 0$.



“Ordered” Fields

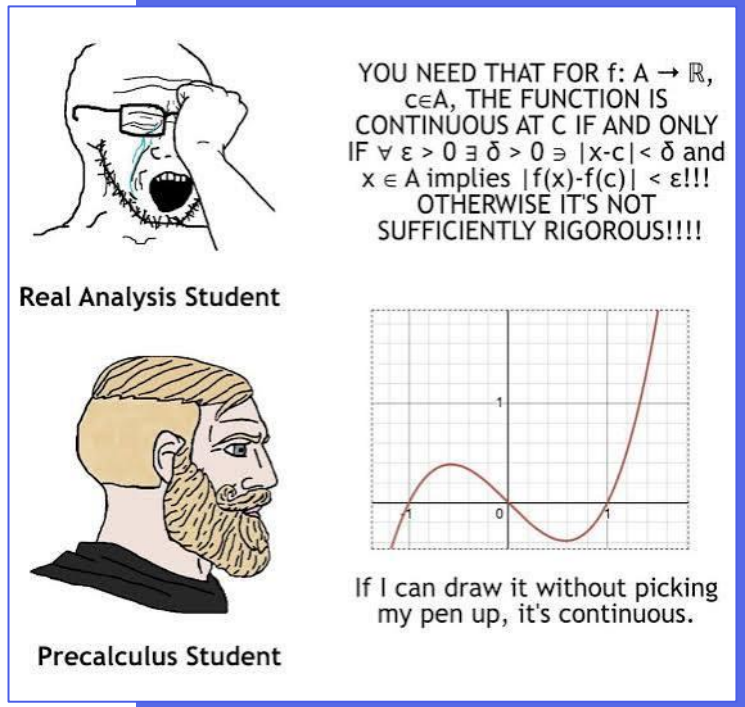
An ordered field consists of a field F together with an ordering < on that field that satisfies the following axioms:—

1. The ordering satisfies the Trichotomy Law.

2. The ordering is transitive.

3. if x, y and z are elements of F and if $x < y$ then $x + z < y + z$.

4. if x and y are elements of F which satisfy $0 < x$ and $0 < y$ then $0 < xy$.

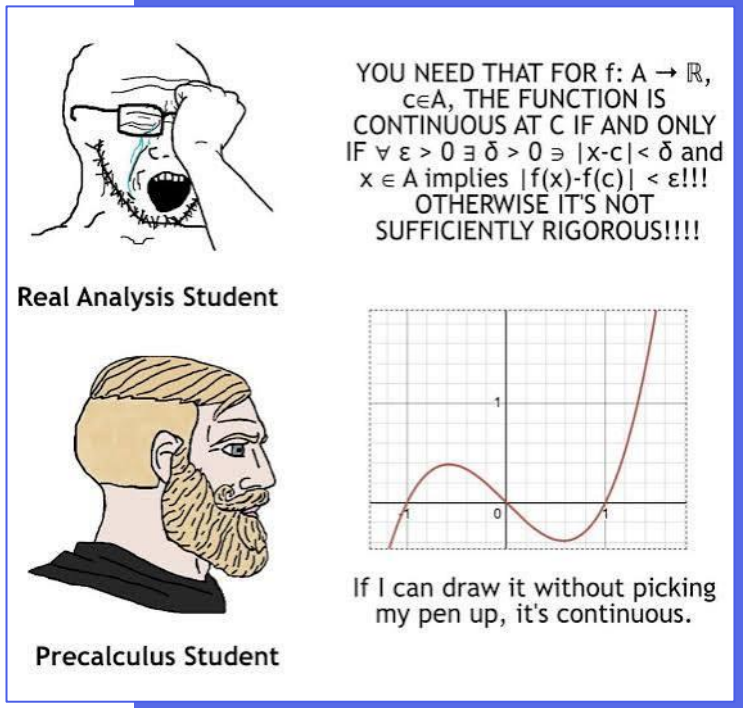


“Ordered” Fields

We can write $x > y$ in cases where $y < x$. we can write $x \leq y$ in cases where either $x = y$ or $x < y$. We can write $x \geq y$ in cases where either $x = y$ or $y < x$.

Okay, it seems that you have already known a lot about Ordered Fields. Here is an example:

Let $Q(\sqrt{2})$ denote the set of all numbers that can be represented in the form $b + c\sqrt{2}$, where b and c are rational numbers. Is it an ordered field? Can you prove it?



Upper Bound



What is upper bound? And what can it do? (that is, why introducing such a concept?)

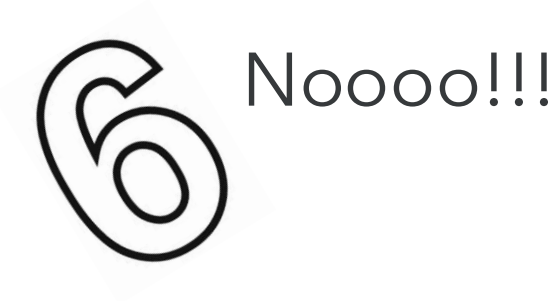
Upper Bound



Let S be a subset of an ordered field $\mathbb{F}$. An element u of $\mathbb{F}$ is said to be an upper bound of the set S if $x \leq u$ for all $x \in S$. The set S is said to be bounded above if such an upper bound exist

And the definition of least upper bound:

Let F be an ordered field, and let S be some subset of $\mathbb{F}$ which is bounded above. An element s of $\mathbb{F}$ is said to be the least upper bound (or supremum) of S (denoted by sup S) if s is an upper bound of S and $s \leq u$ for all upper bounds u of S.



Upper Bound

What can least upper bound do? Let me introduce the least upper bound property:



Given any non-empty subset S of $\mathbb{F}$ that is bounded above, there exists an element sup S of F that is the least upper bound for the set S.

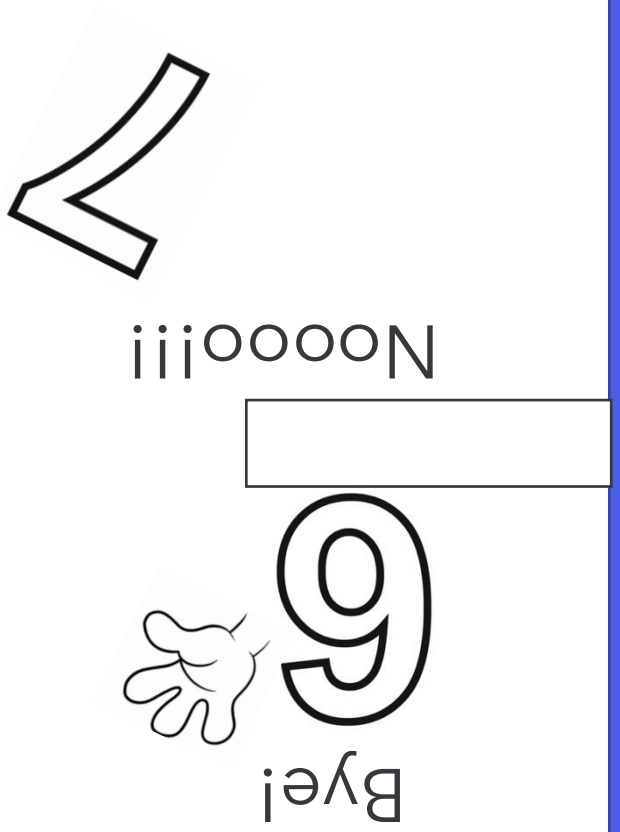
★ A Dedekind-complete ordered field $\mathbb{F}$ is an ordered field which has the Least Upper Bound Property.

Lower Bound

It is basically the same thing, but as a responsible club leader, I will still introduce it.

Let S be a subset of an ordered field $\mathbb{F}$. A lower bound of S is an element l of $\mathbb{F}$ with the property that $l \leq x$ for all $x \in S$. The set S is said to be bounded below if such a lower bound exists. A greatest lower bound (or infimum) for a set S is a lower bound for that set that is greater than every other lower bound for that set. The greatest lower bound of the set S (if it exists) is denoted by inf S.

Let F be a Dedekind-complete ordered field. Then, given any non-empty subset S of $\mathbb{F}$ that is bounded below, there exists a greatest lower bound (or infimum) inf S for the set S. Indeed $\inf S = -\sup\{x \in R : -x \in S\}$.



Bounded Sets of Real Numbers

The set $\mathbb{R}$ of real numbers, with its usual ordering algebraic operations, constitutes a Dedekind-complete ordered field. Thus every non-empty subset S of $\mathbb{R}$ that is bounded above has a least upper bound (or supremum) sup S, and every non-empty subset S of $\mathbb{R}$ that is bounded below has a greatest lower bound (or infimum) inf S.



But what the F*** is this?

What does “with its usual ordering algebraic operations” mean?

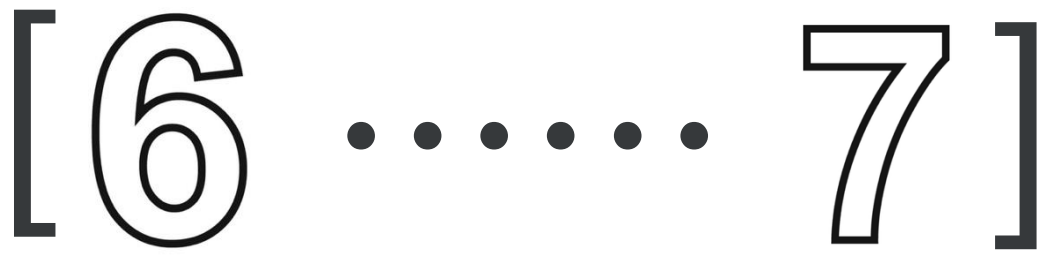
Why is the set $\mathbb{R}$ of real numbers a Dedekind-complete ordered field.

…….

Bounded Sets of Real Numbers

Let S be a non-empty subset of the real numbers that is bounded (both above and below). Then the closed interval [inf S, sup S] is the smallest closed interval in the set $\mathbb{R}$ of real numbers that contains the set S. Indeed if S ⊂[a, b], where a and b are real numbers satisfying a ≤b, then a ≤inf S ≤sup S ≤b, and therefore

$$S \subset [\inf S, \sup S] \subset [a, b].$$



Absolute Values of Real Numbers

We all know how absolute value is defined:

$$|x| = \begin{cases} x & \text{if } x \geq 0; \\ -x & \text{if } x < 0; \end{cases}$$

We have:

1.Let u and v be real numbers. Then |u + v |≤|u|+ |v| and |uv|= |u||v|.

And

2. Let u and v be real numbers. Then| |u|−|v| |≤|u−v|.

2 The polynomial $f(x)$ is defined by

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0\,,$$

where $n \geqslant 2$ and the coefficients $a_0, \ldots, a_{n-1}$ are integers, with $a_0 \neq 0$. Suppose that the equation $f(x) = 0$ has a rational root p/q, where p and q are integers with no common factor greater than 1, and $q > 0$. By considering $q^{n-1}f(p/q)$, find the value of q and deduce that any rational root of the equation $f(x) = 0$ must be an integer.

(i) Show that the nth root of 2 is irrational for $n \geqslant 2$.

(ii) Show that the cubic equation

$$x^3 - x + 1 = 0$$

has no rational roots.

(iii) Show that the polynomial equation

$$x^n - 5x + 7 = 0$$

has no rational roots for $n \geqslant 2$.

THANK YOU SO MUCH

Bill

